Bureau
International des
Poids et
Mesures

# Le système international d'unités
# The International System of Units

9 édition 2019

Annexe 2
Annex 2

## Valeurs recommandées des fréquences étalons destinées à la mise en pratique de la définition du mètre et aux représentations secondaires de la seconde

## Recommended values of standard frequencies for applications including the practical realization of the metre and secondary representations of the second

Partie 2
Part 2

## Mercury 199 Atom ()

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CCL-CCTF Frequency Standards Working Group

**Bureau International**
**des Poids et Mesures**

# Recommended values of standard frequencies for applications including the practical realization of the metre and secondary representations of the second
# Part 2: Mercury 199 Atom ()

**CCL-CCTF Frequency Standards Working Group**

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Appendix 2

# 1. Recommended value CCTF 22e réunion (2020) of the frequency in the CIPM List of Frequencies

$f\left(^{199}\mathrm{Hg}\right) = 1\ 128\ 575\ 290\ 808\ 154.32\ \mathrm{Hz}$

equivalent to

$\lambda\left(^{199}\mathrm{Hg}\right) = 265\ 637\ 977.759\ 838\ 97\ \mathrm{fm},$

with a relative standard uncertainty of $2.4 \times 10^{-16}$.

This radiation was endorsed by the CCTF as a secondary representation of the definition of the second CCTF 21e réunion (2017).

# 2. Method to establish the recommended value

A global adjustment of all measurements of frequency ratios published in peer-reviewed publications and available to the CCL-CCTF WGFS was carried out following the methods presented in [3] to [7] .

This adjustment determines the frequency of 14 transitions (see Figure 1, see p. 5) which are either already adopted as secondary representations of the second [7] or considered as candidates for SRS. It took into account 105 measurements, including 33 frequency ratios and 72 absolute frequency measurements (i.e. ratios to the $^{133}$Cs frequency). A total of 483 correlations between these input measurements were estimated and considered in the adjustment. More details on the input data and the processing are provided at https://webtai.bipm.org/ftp/pub/tai/publication/wgfs/Adjustment_2021.html . The recommended value is the direct result of the adjustment, rounded as deemed adequate with respect to the recommended uncertainty.

While the results are from a global adjustment, it is of interest to note (see Figure 1, see p. 5) that the $^{199}$Hg transition is involved in 2 measurements relative to $^{133}$Cs, and in 4 frequency ratios, 3 of which are with optical transitions.

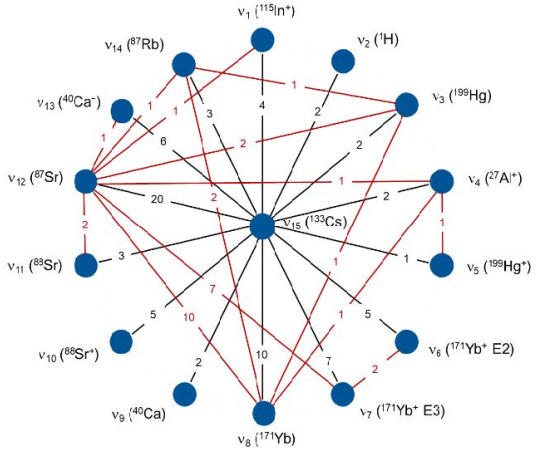



**Figure 1 — Representation of the 105 measurements linking 14 transitions on the circle and $^{133}$Cs at the center.**

## $^{199}$Hg neutral atom, $6s^2\,{}^1S_0 - 6s6p\,{}^3P_0$ unperturbed optical transition

# References


[1] CCTF 22$^{e}$ réunion (2020) BIPM (2020) *22nd meeting of the CCTF*. (Paris: International Bureau of Weights and Measures, Bureau international des poids et mesures) .

[2] CCTF 21$^{e}$ réunion (2017) BIPM (2017) *21st meeting of the CCTF*. (Paris: International Bureau of Weights and Measures, Bureau international des poids et mesures) .

[3] H S Margolis and P Gill (2015) Least-squares analysis of clock frequency comparison data to deduce optimized frequency and frequency ratio values. *Metrologia.* ***52*** *(5) 628–634. https://doi.org/10.1088/0026-1394/52/5/628.*

[4] L Robertsson (2016) On the evaluation of ultra-high-precision frequency ratio measurements: examining closed loops in a graph theory framework. *Metrologia.* ***53*** *(6) 1272–1280. https://doi.org/10.1088/0026-1394/53/6/1272.*

[5] G. Panfilo, communication to the CCL-CCTF WGFS. A new implementation of [4] was realized in MatLab at the BIPM (2020)

[6] Ch. Oates, communication to the CCL-CCTF WGFS. An independent program was developed in Mathematica at NIST (2017)

[7] Fritz Riehle, Patrick Gill, Felicitas Arias and Lennart Robertsson (2018) The CIPM list of recommended frequency standard values: guidelines and procedures. *Metrologia.* ***55*** *(2) 188–200. https://doi.org/10.1088/1681-7575/aaa302.*